

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination - December - 2019

Course: B. Pharm.

Subject with Subject Code: Medicinal Chemistry-II (BP501T)

Date: 16/12/2019

Semester: V

Marks: 75

Duration: 3hrs

- Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. 1 Choose the correct alternative.

1. Which of the following is not a second generation antihistaminic

- A. Cetirizine
B. Cyclizine
C. Loratidine
D. Fexofenadine

2. Which of the following actions is not caused by H_1 receptor

- A. Vasoconstriction
B. Gastric acid secretion
C. Increase permeability
D. Bronchoconstriction

3. Gynaecomastia may be associated with administration of

- A. Ranitidine
B. Cimetidine
C. Terfenadine
D. Omeprazole

4. Rosiglitazone mechanism of action is

- A. Acts as PPAR gamma agonist
B. Inhibitor of alpha glycosidase
C. Acts as amylin analogue
D. Acts as dipeptidyl peptidase inhibitor

5. Which of the following is long acting insulin?

- A. Lispro
B. Aspart
C. Glargine
D. Glulisine

6. Mechanism of action lovastatin is by

- A. Competitive inhibition of rate limiting step in cholesterol synthesis
B. Bile acid sequestration
C. Activate lipoprotein lipase
D. Inhibits lipolysis and triglyceride

7. Which of the following is long acting corticosteroid?

- A. Triamcinolone
B. Betamethasone
C. Hydrocortisone
D. Prednisolone

8. Steroid with equal mineralocorticoid and glucocorticoid activity is

- A. Betamethasone
B. Dexamethasone
C. Hydrocortisone
D. Beclomethasone

9. Which of the following are Alkylating agents except?

- A. Busulfan
- B. Carmustine
- C. Decarbazine
- D. Etoposide

10. Anticoagulant with both *in vitro* and *in vivo* activity is

- A. Heparin
- B. Warfarin
- C. Apixaban
- D. Dabigatran

11. Why anticancer drug thalidomide was banned in 1960?

- A. It inhibits the RNA synthesis
- B. Teratogenic effect
- C. Causes breast carcinoma
- D. Cross linking of DNA

12. Why loop diuretics are contraindicated with aminoglycosides?

- A. Nephrotoxicity
- B. Neurotoxicity
- C. Ototoxicity
- D. Phototoxicity

13. Which of the following drug is not used in Prinzmetal angina?

- A. Propranolol
- B. Verapamil
- C. Nitrates
- D. Isosorbidedinitrate

14. Cough is a adverse reaction seen with intake of _____

- A. Thiazide
- B. Ifedipine
- C. Enalapril
- D. Prazosin

15. Which of the following ACE inhibitor is not a prodrug?

- A. Fosinopril
- B. Enalapril
- C. Ramipril
- D. Lisinopril

16. Which drug prevents peripheral conversion of T_4 to T_3 ?

- A. Propylthiouracil
- B. Propranolol
- C. Iodides
- D. Diltiazem

17. Digitalis toxicity can cause

- A. Hyperkalemia
- B. Nausea
- C. Arrhythmias
- D. All of the above

18. Which of the following anti arrhythmic drug decreases the action potential duration in purkinje fibres?

- A. Quinidine
- B. Flecainide
- C. Amiodarone
- D. Lidocaine

19. Sildenafil acts by inhibiting _____.

- A. Phosphodiesterase-2
- B. Phosphodiesterase-5
- C. Adenyl cyclase
- D. Guanyl cyclase

20. Digoxin acts by inhibiting _____.

- A. $\text{Na}^+ - \text{K}^+$ ATPase B. $\text{H}^+ - \text{K}^+$ ATPase
C. $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ channel D. $\text{Na}^+ - \text{K}^+$ ATPase

Q. 2 Answer any two of the following questions.

- A. Classify antihistamines and add a note on general SAR of classical antihistaminic. Give the mechanism of action for H_2 receptor antagonist as well as sketch out synthetic route for Cimetidine.
- B. What are diuretics? Write classification of diuretics with example from each category. Briefly discuss about Thiazides.
- C. Classify oral anti-hyperglycemic agents with suitable examples and structures. Add a note on sulfonylureas anti-hyperglycemic agents.

Q. 3 Answer any seven of the following questions.

- A. Define and classify anticancer agents. Write a note on alkylating agents.
- B. Write synthesis for (Any two)
- Furosemide
 - Diphenhydramine
 - Methotrexate
 - Tolbutamide
- C. Write a note on ACE inhibitors.
- D. Explain important structural features and mode of action of local anesthetics.
- E. Write a note on Thyroid hormones and antithyroid drugs.
- F. Comment on Oral Contraceptives.
- G. Write a note on Proton Pump Inhibitor.
- H. Classify and explain anticoagulant.
- I. What are cardiotonics? Explain in detail.

Paper End



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: V

Subject Name: Medicinal Chemistry -II

Subject Code: BP501T

Max Marks: 75

Date: 13/07/2022

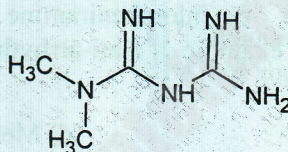
Duration: 3.45 Hr.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

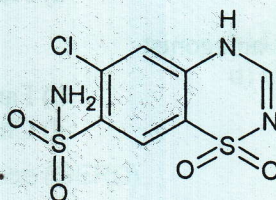
Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

i) Identify the structure



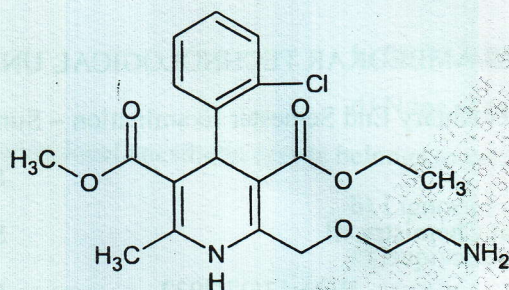
- a) Acarbose b) Metformin c) Glipizide d) Nateglinide.

ii) Identify the structure

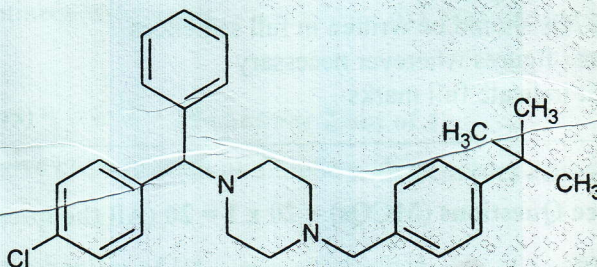


- a) Triamterene b) Chlorthiazide c) Bumetanide, d) Spironolactone,.

iii) Identify the structure



- iv) Identify the structure



- v) Which of the following is NOT example of Anti-arrhythmic drug

- b) Lidocaine
d) Mexiletine

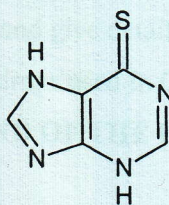
- vi) Which of the following anticancer agent oxidatively damage the cell DNA by binding to metal ion like iron

- b) Cisplatin,
d) Fluorouracil,

- vii) Following is an example of H1 antagonist

- b) Famotidine
d) Chlorpheniramine

- viii) Predict the following structure



- b) Thioguanine
d) Fluorouracil

ix) Following drug competitively inhibits aldosterone dependant sodium potassium exchange channels in the distal convoluted tubule

- a) Spironolactone
- c) Mannitol

- b) Furosemide
- d) Bumetanide,

x) Testosterone is

- a) 17 β -Hydroxyandrost-4-en-3-on
- b) 17,21-Dihydroxypregn-4-ene-3,20-dione
- c) 3 α -Hydroxy-5 α -androstan-17-one
- d) 17, hydroxypregn-4-ene-3,13-dione,

xi) Stomach discomfort, gas, bloating are the side effects of

- a) Pioglitazone
- c) Acarbose

- b) Rosiglitazone
- d) Chlorpropamide

xii) Which of the following is NOT oral contraceptive agent

- a) Mifepristone
- c) Levonorgestrol

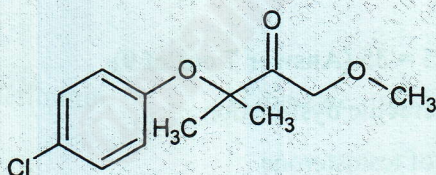
- b) Norgestril
- d) Prednisolone

xiii) The following sulphonamide used in the treatment of diabetes

- a) Acarbose
- c) Nateglinide.

- b) Tolbutamide
- d) All of above

xiv) Identify the structure



- a) Clofibrate
- c) Cholesteramine.

- b) Lovastatin
- d) Cholestipol

xv) Use of cimetidine is

- a) Gastroesophageal reflux disease (GERD)
- c) Allergy

- b) Bloating
- d) Sneezing

xvi) find out the antimetabolite shares structural similarity with folic acid

- a) Mercaptopurine
- c) Fluorouracil,

- b) Thioguanine,
- d) Methotrexate

xvii) Which of following is NOT prodrug

- a) Lisinopril
- c) Cortisone

- b) Clofibrade
- d) Azathioprine

xviii) Onset of action of nitroglycerin is

a) 05-06 mints

c) 05-10 mints

b) 01-03 mints

d) None of these

xix) which of the following local anesthetic agents belongs to ester group

a) Procain

c) Mepivacaine

b) Lignocain

d) Bupivacain

xx) The cisplatin can be used as

a) Alkylating antineoplastic agent

c) Antibiotic antineoplastic agent

b) Antimetabolite antineoplastic agent

d) Cytotoxic antineoplastic agent

Q. 2. Long Answers) = 2 x 10 = 20 (Answer 2 out of 3)

i) Define diuretics, give classification of diuretic with examples and explain mechanism of action and synthesis of Furosemide.

ii) Compare action mechanism of H1 antagonist and H2 antagonist explain advantages of H2 antagonist over H1 antagonist

iii) Classify antidiabetic agent with example, explain mechanism of action of Repaglinide and give synthesis of Tolbutamide

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

i) Write a note on drugs for erectile dysfunction

ii) Explain Stereochemistry of testosterone

iii) Give synthesis of Promethazine hydrochloride

iv) Write synthesis of Warfarin

v) Write a note on SAR of Local anaesthetics

vi) Classify antineoplastic agents,

vii) Discuss pharmacological uses of thyroid and antithyroid drug

viii) Define and classify corticosteroid and give detailed account of prednisolone

ix) Elaborate the action mechanism of drugs used in Congestive Heart Failure

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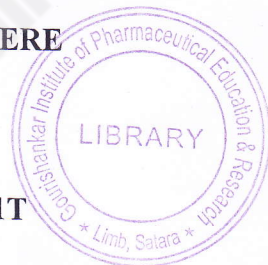
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 08/02/2023

Course: B. Pharmacy
Subject Name: Medicinal Chemistry-II
Max Marks: 75

Sem: V
Subject Code: BP501T
Duration: 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define diuretics with suitable examples.
- ii) Discuss the MOA of calcium channel blocker.
- iii) Classify anti-arrhythmic agents?
- iv) Discuss the MOA of Heparin.
- v) Draw the structure of Cortisone and Hydrocortisone.
- vi) How antimetabolites act as anticancer agents?
- vii) Discuss the role of nitric oxide in human body?
- viii) Write the scheme of synthesis of Benzocaine.
- ix) Explain the diseases of Thyroid gland with suitable example.
- x) Write a note on oral contraceptives.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

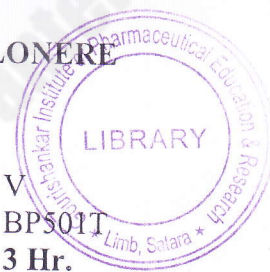
- i) Classify the antihypertensive agents? Discuss in detail about ACE inhibitors and Angiotensin II Receptor Blocker along with SAR.
- ii) Give the classification of Antidiabetic agents with structures from each class. Discuss the SAR of sulfonylureas and draw the synthesis of Tolbutamide.
- iii) Classify antihistaminic agents by giving examples of each class. Explain in detail H1 antagonists and outline the synthesis of Diphenhydramine hydrochloride.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain the chemistry and MOA of cardiac glycosides.
- ii) Discuss in detail about loop diuretics and thiazide diuretics.
- iii) Justify the sentence "Nitro vasodilator act as anti-anginal agent" with suitable example.
- iv) Classify antihyperlipidemic agents with suitable example. Explain MOA of HMG-CoA Reductase Inhibitors.
- v) Classify local anesthetics with suitable example and explain SAR of local anesthetics.
- vi) Discuss in detail about testosterone sex hormone with its structure, metabolism and SAR.
- vii) Classify anticancer agent with suitable examples and explain Alkylating agents.
- viii) Give an account of anticoagulants. Write the synthesis of warfarin.
- ix) Discuss proton pump inhibitors along with MOA.

Course : B. Pharmacy
Subject Name : Medicinal Chemistry- II
Max Marks : 75

Sem: V
Subject Code : BP501T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Write mechanism of action of chlorthiazide.
- ii) What is arrhythmia? Classify Anti-arrhythmic Drugs.
- iii) What are Antihistaminic agents? Classify them.
- iv) What are oral contraceptives?
- v) How antimetabolites will act as an anticancer agent?
- vi) Classify antianginal drugs.
- vii) Write synthetic scheme for warfarin.
- viii) How Digoxin will help to cure congestive heart failure ?
- ix) Write synthetic scheme for tolbutamide.
- x) Brief about stereochemistry of steroids.

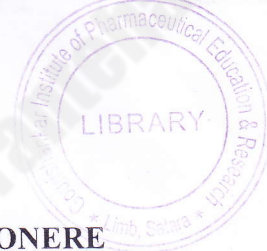
Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain mechanism of action and SAR of local anesthetic agents. Give the synthetic route of dibucaine.
- ii) Classify the oral hypoglycemic agents with examples and explain biguanides and thiazolidinediones in details.
- iii) What are antihypertensive drugs? Classify them with examples and add a note on loop diuretics.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Note on Anti-hyperlipidemic agents.
- ii) What are steroids? Write a note on corticosteroids.
- iii) What is hyperthyroidism. Write a note on antithyroid drugs.
- iv) Explain proton pump inhibitors with examples.
- v) Explain alkylating agents as antineoplastic drugs.
- vi) Write synthesis of diphenhydramine and promethazine.
- vii) Classify anti anginal drugs.
- viii) Note on calcium channel blockers.
- ix) Explain the mechanism of action of vasodilators with examples.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2023 Date:

Course : B. Pharmacy

Sem: V

Subject Name : Medicinal Chemistry-II

Subject Code : BP501T

Max Marks : 75

Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10x 2) = 20

- i) Write a note on histamine and its receptor.
- ii) Give scheme of Synthesis for diphenhydramine HCl.
- iii) Discuss mechanism of action of Acetazolamide.
- iv) Write structure and uses of any two amino alkyl ether derivatives as antihistaminic agents.
- v) Write structure and uses of Synthetic estrogens.
- vi) Write MOA of antimetabolites as antineoplastic agent's.
- vii) Give metabolism of steroids.
- viii) Write a note on adrenocorticoids.
- ix) Write a note on drugs used for erectile dysfunction.
- x) Write MOA of K⁺ sparing diuretics.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Classify antihistaminic agents along with example of each class. Explain in detail H₁-antihistaminics and outline the synthesis of Promethazine HCl.
- ii) What are diuretics? Write its classification along with its structure. Outline the synthesis of Furosemide and write its uses.

- iii) Classify oral antihyperglycemic agents with suitable example and structures. Add a note on Sulfonyl urea as antihyperglycemic agents.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) What are lipid lowering agent? Give structure and uses of Lovastatin, Clofibrate and Cholestyramine.
- ii) Classify Antiarrhythmic agents with example. Give Synthesis of Disopyramide phosphate.
- iii) Write a note on cardiac glycosides used in the treatment of CHF.
- iv) Write a note on Anticoagulant. Give scheme of synthesis for warfarin.
- v) Explain carbonic anhydrase inhibitors and Give Synthetic scheme for Acetazolamide.
- vi) Write a note on steroids.
- vii) What are antihypertensive agent? Explain with example.
- viii) What are local anaesthetics? Give their SAR.
- ix) What are lipid lowering agent? Give structure and uses of Lovastatin, Clofibrate and Cholestyramine.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Summer Examination – 2024

Course: B. Pharmacy

Subject Name: Medicinal Chemistry II

Max Marks: 75

Date: 13-06-2024

Semester: V

Subject Code: BP501T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10X2) =20

- i) Write mechanism of action and medicinal applications of Ranitidine.
- ii) Write synthesis of Warfarin.
- iii) Explain gastric proton pump inhibitors.
- iv) What is arrhythmia? Classify anti-arrhythmic agents.
- v) Give synthetic scheme of Benzocaine.
- vi) What are oral contraceptives?
- vii) Explain thiazolidinediones as hypoglycemic agents.
- viii) Explain stereochemistry of steroids.
- ix) Discuss about drugs used for erectile dysfunction.
- x) Write a note on corticosteroids.

Q.2. Long Answers (Answer 2 out of 3) (10X2) =20

- i) Define Diuretics. Classify diuretics with examples and write mechanism of action and medicinal applications of drugs belonging to class thiazides.
- ii) Classify oral hypoglycemic agents with suitable examples. Comment in detail on sulphonyl ureas. Draw synthetic route for tolbutamide.
- iii) Classify antihistaminic agents with examples and explain in detail H₁-antihistaminics and outline the synthesis of Diphenhydramine hydrochloride.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Discuss in detail antihyperlipidemic agents.
- ii) Write about coagulant agents.
- iii) Discuss about thyroid and anti-thyroid drugs.
- iv) What are sex hormones? Classify estrogens with suitable examples.
- v) Write a note on cardiac glycosides used in the treatment of CHF.
- vi) Classify antianginal agents with suitable examples.
- vii) Give classification of anticancer agents with suitable examples and explain alkylating agents.
- viii) Classify local anaesthetics with suitable examples and explain SAR of local anaesthetics.

ix)	Classify antihypertensive drugs and give synthesis of methyldopate hydrochloride.
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